Communities of the Interested

Redistricting Institution Congruence with Communities of Interest Among Geocoded Public Comments

Spencer Kuhn | Pennsylvania State University | Department of Political Science

Redistricting and Democracy

- Redistricting, a boundary-drawing process undertaken by every state in the United States at least every decade, has increasingly become a critical factor in determining the health of the nation's representative democracy (Grumbach 2023).
- This process has profound downstream consequences for a variety of political outcomes, including district level competitiveness (Jasny et al. 2025; C. Kenny et al. 2023), voter turnout (Fraga 2016), and the creation of majority-minority and minority-influence districts (Barabas and Jerit 2004).
- Redistricting thus provides state governments with immense power to tip the scales towards or away from an electoral landscape that equally enfranchises the voters that make up their constituencies.

Redistricting and Public Opinion

- Despite it's importance, redistricting is nonetheless at odds with traditional democratic pipelines of connecting public opinion to representation and then policy.
- Retrospective voting is complicated by the fact that the constituencies themselves are shifting
- It's often unclear what the public's "vote" on a particular map would be in the absence of a statewide referendum given low information and salience of the issue among the public and the diversity of local community textures
- The likely overall partisan balance resulting from a boundary configuration can be an ideological signal to voters (McCarthy 2022), but very few are likely to have the knowledge and expertise necessary to evaluate the entirety of a district map down to the precinct level (McFarlane 2024)

The Utility of Community

- Voters might instead have a better sense of their own communities and what separates "where they live" from where places become "somewhere else" based on social, cultural, economic, or geographic features
- Under these conditions, the idea of "Communities of Interest" offers a powerful and more direct link between public opinion and redistricting outcomes.

Communities of Interest

- Stephanopoulos (2012) illustrates that a group of people constitute a community of interest if they demonstrate the following three criteria:
  - Sharing a common geographic range
  - Sharing common socioeconomic or cultural interests
  - Subjective identification as members of a cohesive community
- Respecting communities of interest is typically one of several criteria that many boundary-drawing institutions are constitutionally obligated to respect (Webster 2013)
- These criteria are difficult to balance, and some can even be contradictory

Measuring Communities of Interest

- Measuring the extent and frequency of these communities has proven difficult
- Some scholars have used contextual and behavioral characteristics of smaller geographic units within states (Kruse, Gao, and Mayer 2026; Gimpel and Harbridge-Yong 2020; Makse 2012; Syme 2025; Rossiter, Wong, and Delamater 2018), clustering units into cohesive groups
- Others have sought grassroots input from the public (Chen et al. 2022; Chambers et al. 2022; Rambo 2021), building communities from the combined input from the citizenry
- No procedure has emerged as a clear national standard, although state and federal courts have consistently considered and upheld arguments regarding communities of interest (Stephanopoulos 2012)

Democratized Public Testimony

- Attending a public hearing and writing a full essay are less accessible options for many voters
- Submitting a statewide map requires expert-level knowledge and navigation of frontier redistricting software
- Web-based public comment platforms offer a voice for every citizen with broadband access
- Brief questionnaires with no length requirements and free text options allow respondents to narrow their requests to the communities that matter to them most
- Many states have deployed public comment platforms in a variety of contexts, some explicitly requesting information on communities of interest

Theories of Institutional Response

- Scholars have often found that public comment can influence the final text of rules and legislation (Yackee and Yackee 2006; Sahn 2025; Ingrams 2023)
- However, not only are commenters often an unrepresentative sample of the public, but interest groups, particularly business interests, have been found to exact an outsized role in the process of successfully requesting any changes
- Institutionally-collected redistricting testimony may similarly be subject to interest group influences (Rambo 2021; McWhorter et al. 2026)
- Partisan legislators, furthermore, often discount the opinions of constituents holding divergent viewpoints from their own (Butler and Dynes 2016)

Redistricting Testimony Dynamics

- Redistricting is a highly consequential policy undertaken at the state legislative, executive, and judicial levels, dictating representational and democratic dynamics across a state for up to a decade at a time (Grumbach 2023; Fraga 2016)
- States have adopted a variety of map-drawing procedures, each giving partisan legislatures a different level of leeway over determining the final maps based on systems of oversight and stalemate-breaking (McCartan et al. 2026)
- Therefore, institutional response likely depends on an interaction between partisan leeway and the partisanship or geographic nature of commenter requests
- However, the difficult task of synthesizing these comments, alongside their quantity and relative anonymity, likely make them an inefficient medium for partisan interest group efforts

Intertwined Research Questions

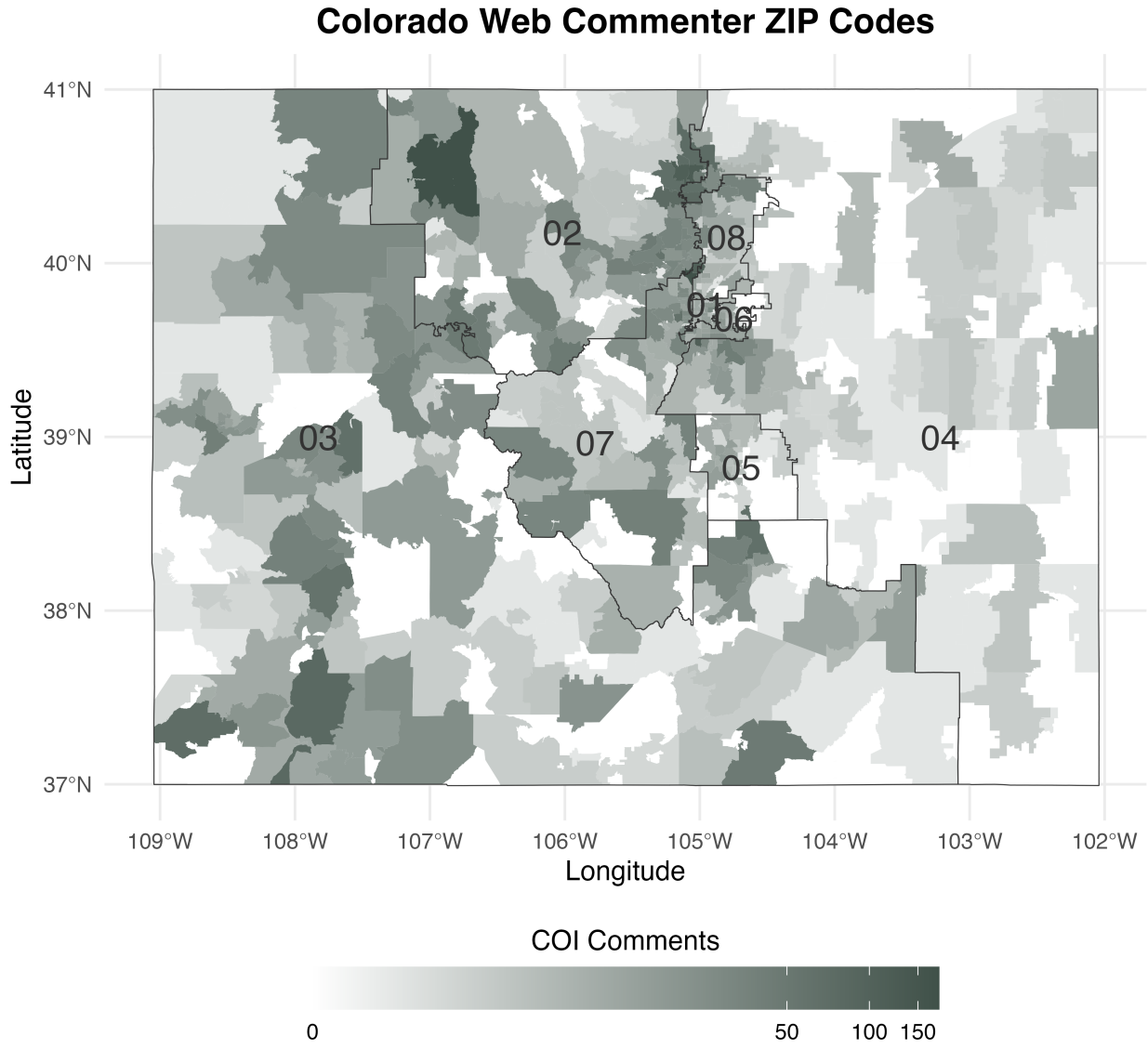
- Descriptively, what can we say about the communities identified by these commenters based on precinct level partisanship, racial makeup, and similarity to baseline political subdivisions?
- Within states, do boundary-drawing institutions more closely adhere to the community of interest requests of public comments as the redistricting process transpires?
- Are certain requests more likely to be honored based on partisanship, racial makeup, or geography?

Methodological Roadmap

1. Collect web comment testimony
2. Screen web comments for mention of COIs
3. Extract locations from web comments
4. Extract grouping and separation requests between locations
5. Match locations to precinct level data
6. Evaluate community network structure and covariate relationships
7. Define request-level adherence for a state's initial, proposed, and final maps
8. Model request-level adherence as a function of community race and partisanship

Data Collection

- Combination of downloads from the Redistricting Data Hub, web scrapers, and FOIA requests to gather thousands from comments from states with sizable testimony submission platforms
- Colorado and Pennsylvania conducted particularly robust web comment collection efforts, and another large corpus of general web comments is available for Georgia
- These three states cover separate U.S. regions and mixtures of legislative control during the 2021-2022 redistricting cycle
- Unique histories and political geographies across the three states offer a diverse set of location types for our LLM extraction pipeline


Colorado Web Commenter ZIP Codes
41°N
40°N
39°N
38°N
37°N
Latitude
109°W
108°W
107°W
106°W
105°W
104°W
103°W
102°W
Longitude
02
08
01
06
03
07
05
04
COI Comments
0
50
100
150

Example Comment

To Whom It May Concern:

I am very concerned about the preliminary State House Map presented to the public. The map as drawn makes little or no sense, discriminates against communities of color, and eliminates competitive districts, helping to demolish our free and fair democratic elections in El Paso County.

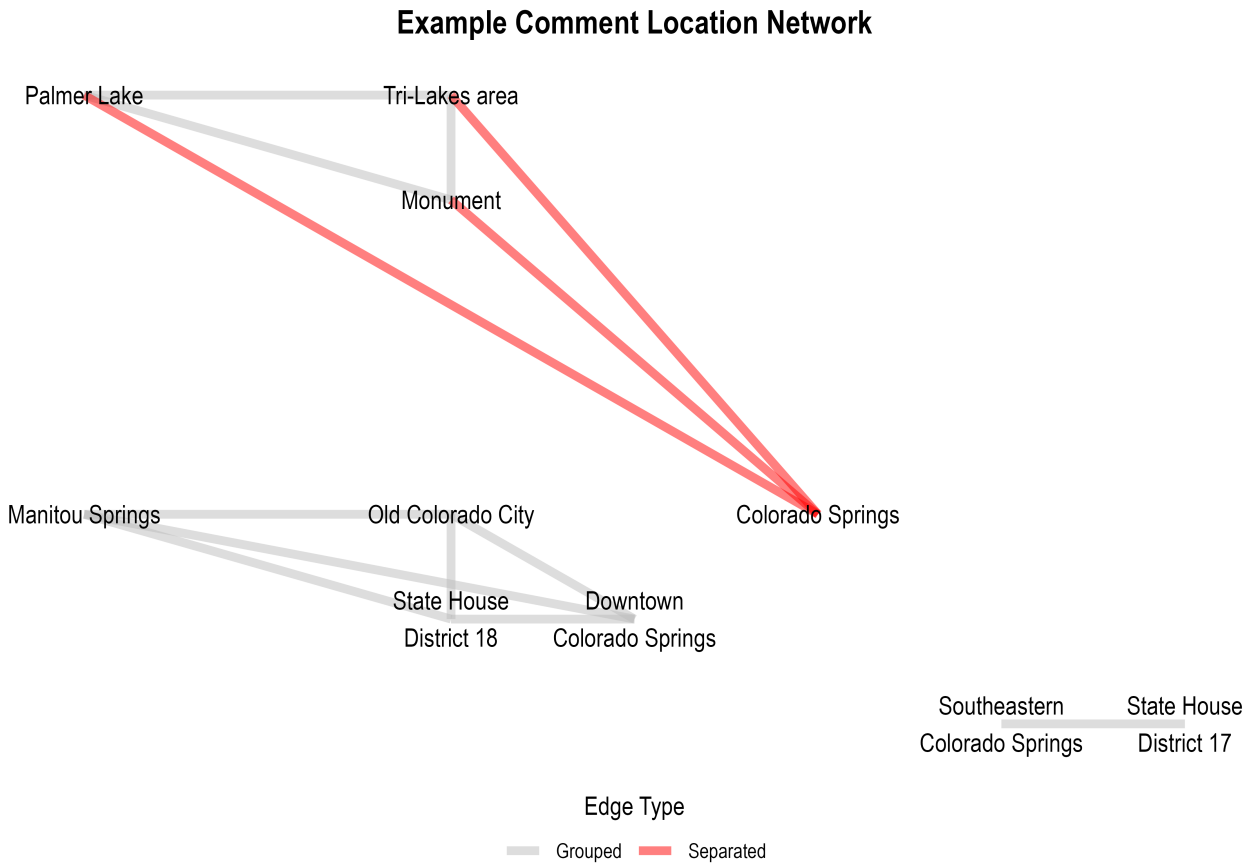
Why are former House Districts 17 and 18 being broken up? Both House Districts are communities of interest. House District 18 is already geographically connected from Manitou Springs to Old Colorado City to downtown Colorado Springs. This area is comprised of older homes with similar interests and needs. House District 17 is another community of interest, serving the needs of Southeastern Colorado Springs. Destroying House District 17 is a disservice to the communities of color that lie within its boundaries. Both of these districts are competitive. Changing them would nullify that fact and certainly make them non-competitive.

Furthermore, I am mystified by your delineation of the boundaries of the proposed House District 49. It comprises only half of the Tri-Lakes area, making absolutely no sense. The Tri-Lakes area is another community of interest and should be kept together as a geographically contiguous district. The Tri-Lakes area of Monument and Palmer Lake has totally different needs from Colorado Springs. Colorado Springs should definitely not be apportioned to the proposed House District 49.

I hope that in the future this State House Map contains competitive districts and accommodates communities of interest, not blatant political opportunism and gerrymandering. This is still a representative democracy, a republic, and public officials such as yourselves need to act in a forthright fashion and champion the democratic ideals of our republic.

Respectfully submitted,

Charlotte gagne


Example Comment Location Network
Palmer Lake
Tri-Lakes area
Monument
Manitou Springs
Old Colorado City
Colorado Springs
State House
District 18
Downtown
Colorado Springs
Southeastern
Colorado Springs
State House
District 17
Edge Type
Grouped
Separated

LLM Annotation of Political Texts

- LLM Annotation has emerged as a defensible method of extracting information from political texts, with a variety of approaches suggested based on experiments with major models (Ornstein, Blasingame, and Truscott 2025; Benoit et al. 2026; Liu and Shi 2025; Heseltine and Clemm von Hohenberg 2024)
- Replicability is of primary concern when using LLMs given their black box nature, which is why Barrie, Palmer, and Spirling advocate for the use of open source and open-weight models (2025)
- Our approach therefore makes use of `ellmer`, a recently-developed R package for structured data extraction from free text (Wickham et al. 2025), as well as Mistral AI's open source mistral-large-2512 model

Ellmer Pipeline Location Extraction

- We have developed a complex prompting architecture to extract named locations from web comments, including landmarks, neighborhoods, municipalities, counties, school districts, legislative districts, and regions
- Location subareas, including cardinal directions and other distinctions, are extracted as well
- Prompting is designed to heavily restrict formatting of location names and avoid the extraction of extraneous locations mentioned purely for context or justification
- Extracted locations are compared against a ground truth set of 50 hand-labeled comments for each state, with almost 600 location mentions across Colorado, Georgia, and Pennsylvania with which to measure recall, precision, and overall accuracy

Ellmer Pipeline Request Extraction

- The second stage of our `ellmer` pipeline evaluates the commenter's requested relationships between mentioned locations in the first stage
- Grouping requests and separation requests are distinct, requiring different means of downstream adherence evaluation
- Extracted requests and their resultant communities are compared against a ground truth set of over 300 location groups

Matching to Voting Precincts

- With the locations and their relationships extracted from each comment, commenter COIs can be constructed by matching locations with 2020 voting precincts
- Precincts for municipalities, counties, school districts, and legislative districts can be matched based on boundary files from the `tigris` package in R as well as the matching capabilities of `geomander` (Walker 2025; C. T. Kenny 2023)
- Statewide vernacular regions (i.e. "Happy Valley") can be matched to sets of counties and their corresponding precincts using the same structured LLM outputs from Mistral AI and `ellmer` that were used for location recognition and request assessment
- Additional matching for neighborhoods and landmarks requires the use of a geocoding API

Measuring Request-Level Adherence

- Adherence for all COI commenters can be assessed for any relevant map, including 2020 boundaries, proposed boundaries, and final boundaries for each state's congressional, state senate, and state house districts
  - For grouped requests, adherence is modeled as a population-weighted diversity index measuring the probability that two randomly selected citizens will fall into the same legislative district
  - For separated requests, adherence is a population weighted index of the probability that two individuals, one from each requested set of separated precincts, will fall into different districts
  - These indices are benchmarked against 5000 simulated, nonpartisan district plans from the `alarmdata` R package (C. Kenny et al. 2023) to account for each state's political geography and redistricting criteria

Modeling Approaches

- Commenter-level adherence pre- and post-redistricting for Colorado, Pennsylvania, and Georgia
- Commenter-level adherence based on measures of precinct-level request homophily along partisanship, race, and rurality, as well as commenter traits where available
- Network community detection of aggregate COIs across commenters
- Adherence across commenters based on partisan leeway (McCartan et al. 2026) and pooled commenter traits

Next Steps

- Complete end-to-end modeling pipeline for Colorado, Pennsylvania, and Georgia
- Conduct additional data collection for states with large comment corpora (starting with Michigan, Arizona, New York, and California)
- Model sentiment and approval/disapproval by comment for additional covariates

References I

Barabas, Jason, and Jennifer Jerit. 2004. "Redistricting Principles and Racial Representation." *State Politics & Policy Quarterly* 4 (4): 415–35. https://doi.org/http://www.jstor.org/stable/40421529.

Barrie, Christopher, Alexis Palmer, and Arthur Spirling. 2025. "Replication for Language Models Problems, Principles, and Best Practices for Political Science." *Working Paper*.

Benoit, Kenneth, Scott De Marchi, Conor Laver, Michael Laver, and Jinshuai Ma. 2026. "Using Large Language Models to Analyze Political Texts Through Natural Language Understanding." *American Journal of Political Science*, 1–17. https://doi.org/https://doi.org/10.1111/ajps.70050.

Butler, Daniel, and Adam Dynes. 2016. "How Politicians Discount the Opinions of Constituents with Whom They Disagree." *American Journal of Political Science* 60 (4): 975–89.

Chambers, Erin, Moon Duchin, Ranthony Edmonds, Parker Edwards, J N Matthews, Anthony Pizzimenti, Chanel Richardson, Parker Rule, and Ari Stern. 2022. "Aggregating Community Maps." *Proceedings of the 30th International Conference on Advances in Geographic Information Systems*, no. 27. https://doi.org/10.1145/3557915.3560961.

Chen, Sandra, Samuel Wang, Bernard Grofman, Richard Oberand, Kyle Barnes, and Jonathan Cervas. 2022. "Turning Communities of Interest into a Rigorous Standard for Fair Districting." *Stanford Journal of Civil Rights and Civil Liberties* 18 (1): 101–89.

Fraga, Bernard. 2016. "Redistricting and the Causal Impact of Race on Voter Turnout." *The Journal of Politics* 78 (1): 19–34. https://doi.org/10.1086/683601.

Gimpel, James, and Laurel Harbridge-Yong. 2020. "Conflicting Goals of Redistricting: Do Districts That Maximize Competition Reckon with Communities of Interest?" *Election Law Journal: Rules, Politics, and Policy* 19 (4): 451–71. https://doi.org/https://doi.org/10.1089/elj.2019.0576.

Grumbach, Jacob. 2023. "Laboratories of Democratic Backsliding." *American Political Science Review* 117 (3): 967–84. https://doi.org/10.1017/S0003055422000934.

References II

Heseltine, Michael, and Bernhard Clemm von Hohenberg. 2024. "Large Language Models as a Substitute for Human Experts in Annotating Political Text." *Research & Politics* 11 (1). https://doi.org/10.1177/20531680241236239.

Ingrams, Alex. 2023. "Do Public Comments Make a Difference in Open Rulemaking? Insights from Information Management Using Machine Learning and QCA Analysis." *Government Information Quarterly* 40 (1): 101778. https://doi.org/https://doi.org/10.1016/j.giq.2022.101778.

Jasny, Ethan, Christopher Kenny, Cory McCartan, Tyler Simko, Melissa Wu, Michael Zhao, Aneetej Arora, et al. 2025. "Gerrymandering and Geographic Polarization Have Reduced Electoral Competition." *arXiv*. https://doi.org/https://arxiv.org/abs/2508.15885.

Kenny, Christopher T. 2023. *geomander: Geographic Tools for Studying Gerrymandering*. https://christophertkenny.com/geomander/.

Kenny, Christopher, Cory McCartan, Tyler Simko, Shiro Kuriwaki, and Kosuke Imai. 2023. "Widespread Partisan Gerrymandering Mostly Cancels Nationally, but Reduces Electoral Competition." *Proceedings of the National Academy of Sciences* 120 (25): e2217322120. https://doi.org/10.1073/pnas.2217322120.

Kruse, Jacob, Song Gao, and Kenneth Mayer. 2026. "Modeling Region Affiliation with Fuzzy Membership Based on Spatial and Social Interactions." *Annals of the American Association of Geographers* 116 (2): 289–311. https://doi.org/10.1080/24694452.2025.2551044.

Liu, Menglin, and Ge Shi. 2025. "Enhancing LLM-Based Text Classification in Political Science: Automatic Prompt Optimization and Dynamic Exemplar Selection for Few-Shot Learning." *arXiv*. https://doi.org/https://arxiv.org/abs/2409.01466.

Makse, Todd. 2012. "Defining Communities of Interest in Redistricting Through Initiative Voting." *Election Law Journal: Rules, Politics, and Policy* 11 (4): 503–17. https://doi.org/https://doi.org/10.1089/elj.2011.0144.

McCartan, Cory, Christopher Kenny, Tyler Simko, Emma Ebowe, Michael Zhao, and Kosuke Imai. 2026. "Redistricting Reforms Reduce Gerrymandering by Constraining Partisan Actors." *American Political Science Review*, 1–20. https://doi.org/10.1017/S0003055426101610.

References III

McCarthy, Devin. 2022. "Limits of Partisanship in Citizen Preferences on Redistricting." *Election Law Journal: Rules, Politics, and Policy* 21 (2): 155–70. https://doi.org/10.1089/elj.2020.0709.

McFarlane, Donovan. 2024. "Critical Review of Geographic Illiteracy in the United States: A Multifaceted Analysis." *Journal of Geography, Environment and Earth Science International* 28 (1): 11–25. https://doi.org/10.9734/jgeesi/2024/v28i1741.

McWhorter, Atticus, Caroline Hammond, Nianqiao Phyllis Ju, and Daryl DeFord. 2026. "Redistricting from the Bottom up: Sampling Communities of Interest with Differential Privacy." *arXiv*. https://doi.org/https://arxiv.org/abs/2606.14453.

Ornstein, Joseph, Elise Blasingame, and Jake Truscott. 2025. "How to Train Your Stochastic Parrot: Large Language Models for Political Texts." *Political Science Research and Methods* 13 (2): 264–81. https://doi.org/10.1017/psrm.2024.64.

Rambo, Olivia. 2021. "Defining Communities of Interest: Political Self-Definition in Los Angeles County During the First California Citizen's Redistricting Commission." *GeoJournal* 86: 1545–87. https://doi.org/https://doi.org/10.1007/s10708-020-10141-3.

Rossiter, Kalyn, David Wong, and Paul Delamater. 2018. "Congressional Redistricting: Keeping Communities Together?" *The Professional Geographer* 70 (4): 609–23. https://doi.org/https://doi.org/10.1080/00330124.2018.1443477.

Sahn, Alexander. 2025. "Public Comment and Public Policy." *American Journal of Political Science* 69 (2): 685–700. https://doi.org/https://doi.org/10.1111/ajps.12900.

Stephanopoulos, Nicholas. 2012. "Redistricting and the Territorial Community." *University of Pennsylvania Law Review* 160 (5): 1379.

Syme, James. 2025. "Using Dual Graphs to Model Territorial Communities of Interest." *PREPRINT*. https://doi.org/https://doi.org/10.21203/rs.3.rs-7040797/v1.

Walker, Kyle. 2025. *Tigris: Load Census TIGER/Line Shapefiles*. https://doi.org/10.32614/CRAN.package.tigris.

References IV

Webster, Gerald. 2013. "Reflections on Current Criteria to Evaluate Redistricting Plans." *Political Geography* 32: 3–14. https://doi.org/https://doi.org/10.1016/j.polgeo.2012.10.004.

Wickham, Hadley, Joe Cheng, Aaron Jacobs, Garrick Aden-Buie, and Barret Schloerke. 2025. *Ellmer: Chat with Large Language Models*. https://doi.org/10.32614/CRAN.package.ellmer.

Yackee, Jason Webb, and Susan Webb Yackee. 2006. "A Bias Towards Business? Assessing Interest Group Influence on the u.s. Bureaucracy." *The Journal of Politics* 68 (1): 128–39. https://doi.org/https://doi.org/10.1111/j.1468-2508.2006.00375.x.